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Operating the Mu3e SciFi Detector

Control, Monitoring and Calibration

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Summary

The Mu3e experiment at the Paul Scherrer Institute (PSI) searches for the charged-lepton-flavour-violating decay $\mu^+ \to e^+e^-e^+$. In the Standard Model it proceeds only through a neutrino loop, with a branching ratio of order 10^{-54}: it is effectively background-free, so any observation would be unambiguous evidence of new physics. The current limit, $\mathcal{B} < 10^{-12}$ (SINDRUM, 1988), leaves four orders of magnitude to explore, down to a single-event sensitivity of about 10^{-16}. This requires a continuous $10^8\,\mu$/s beam and the suppression of two backgrounds: internal conversion $\mu \to eee\nu\bar{\nu}$, controlled by momentum resolution, and accidental combinatorics, controlled by timing and vertex resolution. The scintillating-fibre (SciFi) detector provides the timing of the central region, with a target resolution below 200 ps.

2 rings	96	3072	12	24
12 modules each	MuTRiG ASICs	readout channels	front-end boards	HV channels

Project. My project was the SciFi detector's control, monitoring and calibration system, built end to end for the first physics run at PSI in October 2026. SciFi was operated through a single web page (`Scifi-monitor.js`) that had grown organically: it worked, but it was hard to trust and extend, and it lagged behind the mature pixel-tracker tooling. I replaced it with a structured set of Python control scripts that drive the hardware and a new custom web page that a shift operator can rely on, both running inside the MIDAS data-acquisition framework and communicating through its online database (ODB).

Method. The system is built in layers: installation and bring-up at PSI; the operator interface; the power chain (low voltage, high voltage, front-end boards); a configuration and hardware-test hub; the two SiPM calibrations (threshold and breakdown-voltage scans); and the saving of calibration results to three destinations — the live ODB, a local per-board cache and the permanent calibration database (CDB).

Results. The detector — low voltage, high voltage, front-end boards and readout — is installed and working on the experiment back-end; the calibration result-saving path is fully implemented and wired; and the shared control-and-monitoring library now covers SciFi as well as the pixel detector. The main item left is a full end-to-end test on the experiment, including a real calibration run.
Personal contribution. I designed and implemented the page, the hardware-test hub, both calibrations and the saving path, and extended the common library to SciFi. Alongside the software I took part in the on-site detector work — cabling, power distribution, assembly and repairs across the full Mu3e detector — and in the collaboration's weekly reports and meetings.

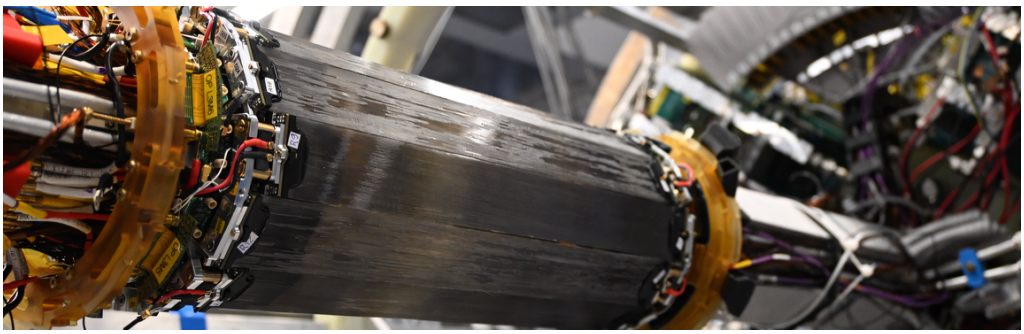


Figure 1. The SciFi detector during installation at PSI.

1 Project Details

1.1 State of the art and objectives

Physics context. Charged lepton flavour violation (cLFV) has never been observed. Neutrinos oscillate, so lepton flavour is violated in the neutral sector, but for charged leptons the Standard Model forbids it to any measurable level: $\mu^+ \to e^+e^-e^+$ needs a neutrino loop and is predicted at $\mathcal{B} \sim 10^{-54}$ [1]. Many extensions of the Standard Model raise this to within reach, making $\mu \to eee$ one of the "golden" muon channels beside $\mu \to e\gamma$ and μ–e conversion. The strongest bound, $\mathcal{B} < 1.0 \times 10^{-12}$ (90% C.L.), is from SINDRUM [2]; Mu3e improves on it by four orders of magnitude using the most intense continuous surface-muon beam in the world [1].

Detector and DAQ. Mu3e reconstructs the three-electron final state with ultra-thin high-voltage monolithic pixel sensors and measures time with two scintillator systems: the SciFi detector in the central region (< 200 ps) and scintillating tiles in the recurl stations (< 50 ps) [3, 4]. SciFi consists of an upstream (US) and a downstream (DS) ring of twelve modules each; its 96 MuTRiG ASICs (32 channels each) are served by twelve front-end boards and read out through the MIDAS framework [5], whose online database holds all configuration and slow-control data and whose web server hosts custom operator pages. One module key — side, ring position, readout end — is used consistently in the ODB, the geometry and the calibration [6].

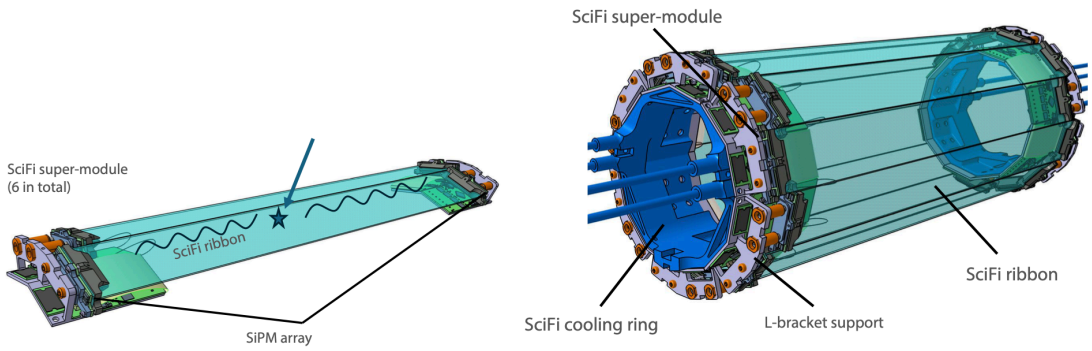


Figure 2. The SciFi detector: a fibre ribbon read out by SiPM arrays at both ends (left) and the full cylinder with cooling rings and L-bracket supports (right).

Starting point. The pixel subdetector already had mature MIDAS integration: operator pages and a structured library of Python control scripts. SciFi relied on one monitoring page that had accreted features, with control logic spread over disconnected scripts and no interface a shift operator could depend on during a run.

Objectives. With the first physics run in October 2026, SciFi needed **(i)** a coherent operator interface, **(ii)** a documented, reproducible calibration whose results reach the CDB, and **(iii)** a control code base shared with, rather than duplicating, the pixel tooling. A dependable control system is a prerequisite for taking SciFi data at all, and unified software lowers the long-term cost of running the experiment.

1.2 Methodology

Environment. Everything targets the MIDAS DAQ: configuration and state live in the ODB, operator interfaces are HTML/JavaScript pages served by MIDAS, and device logic is Python on the front-end machine (Fig. 3). No new physics data was taken; the project builds on the ODB slow-control tree, the existing pixel control library and the diverged SciFi scripts, and on the raw per-channel data of the MuTRiG scans and hardware tests. The fixed hardware layout is the reference against which ODB contents were checked. Development used a local test ODB, then real hardware on the experiment back-end.

Existing vs. new. The pixel pages and scripts were the existing method and served as template: the same web-server/ODB binding pattern and the same *page-first* workflow — mock up the layout, agree the control granularity with the SciFi experts, then write the Python backend. What had to be built was the entire SciFi page and its ODB bindings (the pixel page does not fit the different topology); SciFi coverage in the shared library, factored around subdetector-agnostic abstractions for front-end board, ASIC and channel; a configuration tab

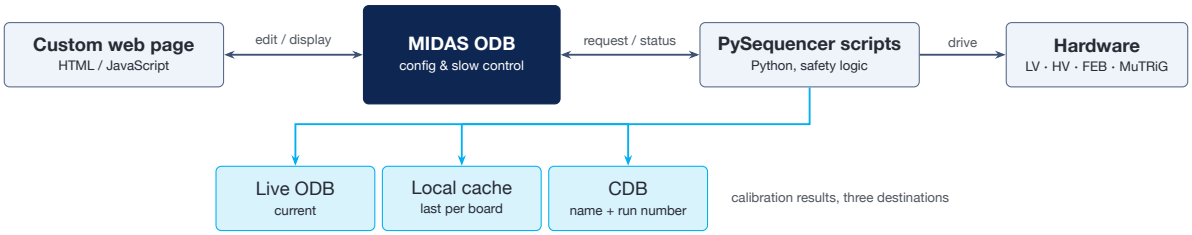


Figure 3. Architecture. All state flows through the ODB. The page never talks to the hardware: it writes requests that a Python script executes, and calibration results fan out to three destinations.

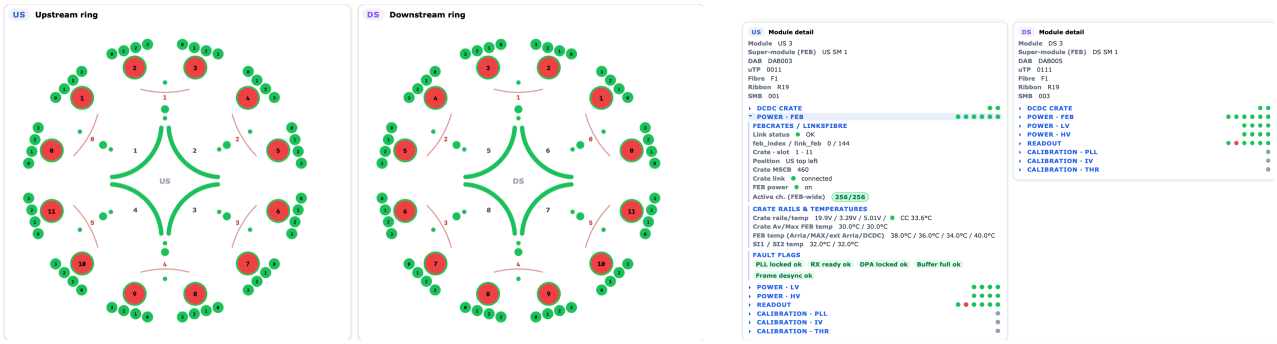


Figure 4. The geometry view of the new operator page. Left: live US and DS rings, with per-module, per-ASIC and per-crate state encoded in colour. Right: drill-down into one module, with collapsible Power, Readout and Calibration sections and fault flags.

that doubles as the hardware-test hub; and two SiPM calibrations on a common lifecycle with a working path to permanent storage.

Interface. A read-only *Geometry* view — a drill-down from super-module to module to the 32-channel ASIC grid, drawn as mirrored US and DS rings — sits beside the editable *Power*, *Readout* and *Calibration* tabs. Every editable value goes to the ODB, from where a front-end process applies it. Safety logic stays in Python and is never duplicated in JavaScript; no action is ever hard-blocked — the page warns rather than refuses.

Configuration and hardware tests. The configuration tab walks the front-end board → module → chip → channel tree, writes to a dedicated ODB subtree, and runs three tests: **PLL** (each board must recover the clock from its data link — the recovered-clock delay histogram is flat when locked and sharply peaked when not), **PRBS** (link integrity) and **TDC** (per-channel test pulse).

Calibration. Two per-board scans share one `prepare` / `execute` / `finalise` lifecycle. The *threshold scan* walks the discriminator threshold on each channel and reads the rate curve to find the 0.5, 1.5 and 2.5 photo-electron crossings; the result is loaded as operating threshold. The *breakdown-voltage scan* differentiates each SiPM's current–voltage curve and fits a Gaussian to the peak, whose mean is the breakdown voltage and therefore sets the operating overvoltage (Fig. 5). Each scan runs independently per board, so any subset of the detector can be re-calibrated. A run is requested from the page via the ODB; the PySequencer launches the script, which opens its own MIDAS connection (explicitly closed on exit).

Saving. Each destination answers one question: the **live ODB** (*what is the calibration right now?*, overwritten each scan), the **local cache** of the latest calibration per board (*what did we last do, day to day?*) and the permanent **CDB** indexed by name and run number (*what was it on a given run?*). The on-disk format is a small ROOT file with trees for metadata, raw scan, results, conditions and channel masks. This path was reconciled with a colleague's parallel work on the same code.

Verification and on-site work. Each branch — power bring-up, hardware tests, calibration scans — was built and checked alone before merging, first on the local test ODB and then on real hardware. The software grew inside the on-site campaigns at PSI, where I also worked on cabling, power distribution, assembly and repairs; the power and readout chains the page represents are largely those I helped bring up by hand.

1.3 Results

All three objectives were met and everything that had to be ready for the October run is in place (Table 1).

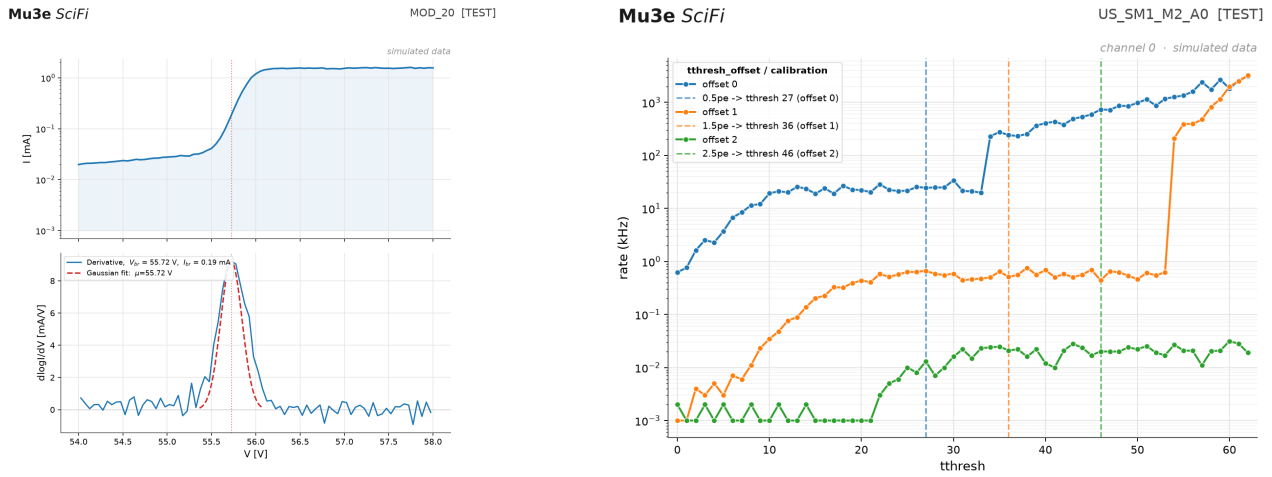


Figure 5. Output of the two calibration scans on simulated data (test ODB). Left: current–voltage curve and Gaussian fit to its derivative, giving the breakdown voltage ($V_{br} = 55.72$ V for this test module). Right: threshold scan.

Objective	Outcome	Status
Operator interface	New page replaces `Scifi-monitor.js`; Power, Readout, Calibration bound to the ODB; Geometry view and readout-status monitoring (rate, run state, buffers) integrated — "power all green" and "data flowing" shown as two distinct conditions	done
Hardware tests	Configuration tab became the test hub, absorbing the Readout tab; PLL, PRBS, TDC return one verdict per board or link	done
Calibration & saving	Both scans run per board and produce the loaded threshold and overvoltage; saving to live ODB, local cache and CDB fully implemented, ROOT layout finalised	done
Shared code	Common library covers SciFi and pixel; duplication removed	done
Experiment status	LV, HV, FEBs and readout up; operated through the new page during installation week at PSI	done
End-to-end test	Page → real calibration run → all three destinations	open

Table 1. Results against the planned objectives.

Expectations and outlook. The delivered scope matches what was agreed with the SciFi group; the one deviation — the configuration tab growing into a full test hub — was in the useful direction. The principal open item is a single end-to-end test on the experiment with a real calibration run. Smaller follow-ups: move the hard-coded 32-cell HV board-to-module map into Python/ODB, extend the online monitoring histograms, and wire SciFi's own `CalibrationFactory` to a real entry point so the scripts no longer need their direct-construction path.

1.4 Contributions as an S3P3 student

I worked within the Mu3e collaboration, SciFi detector group, in coordination with the pixel DAQ developers, who own the shared code and whose tooling was the reference. The system described above is my own work:

- the new custom page end to end — Geometry view, Power / Readout / Calibration tabs, ODB bindings and Python backend — and the design decisions behind it (safety logic in Python only; warn, never block);
- the configuration tab as hardware-test hub, with the PLL, PRBS and TDC flows;
- the threshold and breakdown-voltage calibrations on their shared lifecycle and the full saving path to ODB, cache and CDB, including the ROOT layout and its reconciliation with a colleague's branch;
- the extension of the common control library to SciFi.

Hardware, service and collaboration. Software was half the work. On site at PSI I did SciFi cabling and power-distribution management, mechanical assembly and repairs, and designed small hardware components; much

of my time went to general service work across the full Mu3e detector, including other subdetectors. I reported weekly, took part in subdetector and DAQ meetings, and coordinated continuously with the DAQ team, whose shared code the SciFi tooling extends rather than duplicates.

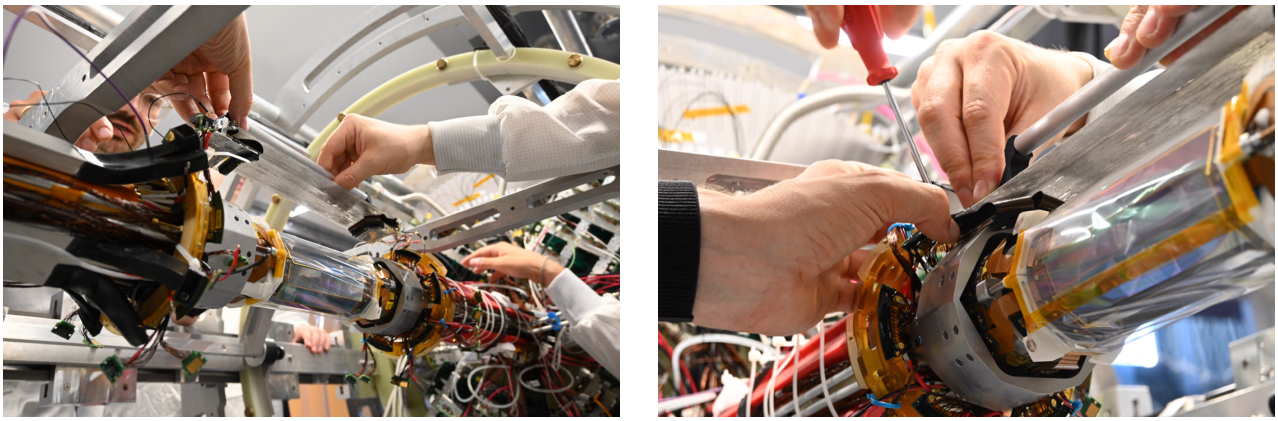


Figure 6. On-site work at PSI: assembly and cabling of the detector.

References

[1] Mu3e Collaboration, “Technical design of the Phase I Mu3e experiment”, arXiv:2009.11690 (2021).

[2] U. Bellgardt *et al.* (SINDRUM Collaboration), “Search for the decay $\mu^+ \to e^+e^+e^-$”, Nucl. Phys. B **299** (1988) 1.

[3] A. Bravar *et al.*, “Development of the Scintillating Fibre Timing Detector for the Mu3e Experiment”, arXiv:2208.09906 (2022).

[4] L. Vigani *et al.*, “The Mu3e detector”, JINST **17** (2022) C05024.

[5] M. Köppel (Mu3e Collaboration), “Mu3e Integration Run 2021”, arXiv:2203.07855 (2022).

[6] Y. Wang (Mu3e Collaboration), “The DAQ system of the Mu3e SciFi detector”, SPS/APS Joint Annual Meeting (2023).